

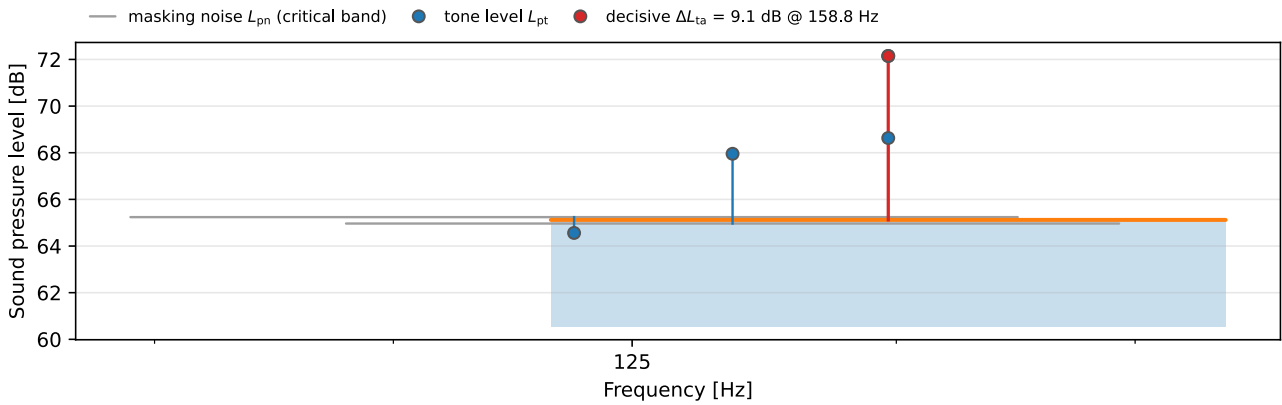
Tonal audibility assessment

ISO 1996-2 tonal audibility by the ISO 1996-2:2017 Annex J engineering method (ISO/PAS 20065:2016); tonal adjustment K per Table J.1.

Source / situation: Combustion engine, steady operation Analysis line spacing Δf 2.7 [Hz]:

Detected tones

Tone f_T [Hz]	Type	L_{pt} [dB]	L_{pn} [dB]	Δf_c [Hz]	ΔL_{ta} [dB]	Present
118.4	Single	64.6	65.2	101	1.3	Present
137.3	Single	68.0	65.0	101	5.0	Present
158.8	Single	68.6	65.1	102	5.5	Present
158.8	FG (3)	72.1	65.1	102	9.1	Present



Tonal audibility $\Delta L_{ta} = 9.1$ dB tonal adjustment
K = 5 dB

Decisive tone $f_T = 158.8$ Hz
Analysis line spacing $\Delta f = 2.7$ Hz
Extended uncertainty $U = 3.3$ dB (90 % coverage)

Result vs requirement: $\Delta L_{ta} = 9.1$ dB, required ≤ 6.0 dB → **FAIL**

A prominent tone is present (decisive $\Delta L_{ta} = 9.1$ dB > 0); the tonal adjustment K = 5 dB (ISO 1996-2:2017 Table J.1) applies.

The tonal audibility $\Delta L_{ta} = L_{pt} - L_{pn} - a_v$ is the amount by which the tone level rises above the masking threshold of the surrounding noise (ISO/PAS 20065:2016, Formula (14)); the decisive value is the largest over the detected tones (Clause 5.3.8).

L_{pn} is the critical-band masking-noise level (Formula (12)), Δf_c the critical bandwidth about the tone (Formula (2)); an FG entry combines tones sharing a critical band (Formula (17)).

Table J.1 assigns K from the mean audibility ΔL , the energy mean of the decisive audibilities of the J assessed spectra (ISO/PAS 20065:2016, Clause 5.3.9, Formula (20)). This fiche assesses a single spectrum, so the K shown follows from its decisive audibility; where several staggered spectra are assessed, rate their mean audibility instead.

Laboratory: Phonometry Reference Laboratory
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■ The results relate only to the tested specimen.
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